

ARITHMETIC AND CALIBRATION

Final RTL evidence | 2026-09-30 | Group 1/3: seven named benches

BENCH / OBSERVED LOG COUNTS

REFERENCE CHECK / COVERAGE BOUNDARY

p2_oracle_tb

1,048,576 input codes; 0 errors

Observed: P2_ORACLE_COMPLETE codes=1048576 errors=0

Identity: dout == input code;
zero flags and 11-cycle completion.
One-slice fixture; not all configurations.

divider_borrow_tb

67,592 operand/config combinations
6 width/unroll fixtures
Observed: DIVIDER_BORROW_COMPLETE exhaustive_checks=67592

Signed floor reference + defining
inequality; zero divisor, busy and hold.
Six small-width parameter fixtures.

cal_weight_reduce_ppa_tb

1,430,848 miter input cases / 7 fixtures
43,758 allocations in 18-slice fixture
Observed: CAL_WEIGHT_REDUCE_PPA_COMPLETE

Bit-exact T/G/R miter against the
previous three-tree RTL + invalid-ID rule.
Shared mathematics: equivalence, not physics.

calibration_physical_tb

2,048 transactions; 18 physical slices

Observed: CALIBRATION_PHYSICAL_COMPLETE

Independent 256-bit integer formula;
code, flags, ID and input capture.
18 slices, reduced 8-cell test geometry.

calibration_recovery_tb

2,048 target/residue transactions
max |error|: calibrated 4; nominal 504 codes
Observed: CALIBRATION_RECOVERY_COMPLETE

Known target -> synthetic residue ->
calibrated and nominal-weight outputs.
Synthetic mismatch + ideal ADC2 quantization.

calibration_fit_tb

128 holdout samples; 1,278 weights
code + flags + ID + 11-cycle latency
Observed: CALIBRATION_FIT_COMPLETE

Frozen external holdout vectors;
code/flags/ID/latency + coefficient readback.
Fitted coefficients loaded, not learned in RTL.

p2_tb

23,049 TB checks; 0 errors

Observed: P2 RESULT: PASS

Golden vectors + floor inequality;
rounding, masks, carry, clipping and link.
T5 full-code sweep is a separate bench.